



Jointly Learning Type-Aware Relations and Inter-Aspect with Graph Convolutional Networks for Aspect Sentiment Analysis

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Abstract

This paper presents a novel approach for aspect-level sentiment analysis by leveraging the relationships between dependent types and aspects. The proposed method involves simplifying the type-aware graph convolutional network and designing a graph convolution module specifically for extracting relations between aspect words. The process begins with constructing an ordinary dependency graph for each sentence using a dependency tree. This graph is then refined by considering syntactic dependencies between context words and aspect-specific words, resulting in an aspect-focused graph. The aspect-focused graph, along with the corresponding embedding matrices, is fed into the aspect-focused GCN to capture the essential aspects and context words. Moreover, an inter-aspect GCN is employed to extract the dependencies between aspect words and other aspect words, utilizing the representations learned by the focused aspect GCN based on the inter-aspect graph. The L-layer of the GCN incorporates a bidirectional attentional mechanism to extract interrelationships, thus enhancing sentiment polarity judgment. Through interactive learning of aspect-specific affective features, the model acquires an understanding of the relationships between important text and aspect words, as well as the relationships among aspect words. Experimental results on five benchmark datasets demonstrate the superior performance of our proposed method compared to state-of-the-art approaches, exhibiting a significant improvement over the regular GCN model.

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1 Introduction

Due to the increasing popularity of social media, sentiment analysis has become an important topic in natural language processing [1, 2]. Unlike traditional sentiment analysis tasks, which operate at the sentence or document level, aspect-level sentiment analysis offers a more granular perspective. Aspect-based sentiment analysis focuses on identifying the sentiment polarity [3, 4]—such as positive, neutral, or negative—associated with a specific aspect within a given sentence. For instance, as depicted in Fig. 1, the sentence “Beverage menu is limited but the wine is the opposite.” highlights two aspects: “Beverage menu” and “wine,” with their respective sentiment polarities being negative and positive. This manuscript concentrates on the sentiment classification aspect of Aspect-Based Sentiment Analysis (ABSA), which involves four primary tasks [5]: aspect term sentiment analysis (ATSA) [6], aspect category sentiment analysis (ACSA) [7], aspect term extraction (ATE) [8], and aspect category extraction (ACE) [9].

To effectively address the ABSA task, capturing the interplay between aspects and their surrounding context is crucial [10, 11]. While attention mechanisms have been extensively integrated into ABSA models [12–14], effectively handling single-aspect terms, their efficacy tends to diminish with phrasal aspect terms and sentences featuring multiple aspects. This is largely due to the increased susceptibility to noise, which can adversely affect the model’s performance. For example, consider the sentence: “The service was excellent, but the coffee was too bitter.” Here, the term “excellent” might incorrectly draw attention to “coffee”, even though it pertains to “service”. As a result, attention mechanisms may struggle to accurately process sentences with multiple aspects and phrasal aspects.

Recent research has shown that the relationship between aspects and context can be examined through syntactic structures [15, 16], as demonstrated by methods like graph attention networks (GATs) and graph convolutional networks (GCNs) [17, 18]. Syntactic dependency trees are often utilized to capture long-distance dependencies between words and their contexts. However, extracting sentiment dependencies across different clauses from these trees poses challenges due to their inherent structure. While syntactic dependency trees effectively establish relationships between words, they struggle to represent more complex relations such as conditional statements, oppositions, or conjunctive joins. As a result, modeling dependencies between multiple aspects becomes problematic.

In this paper we address the above two challenges faced by ABSA, we propose a joint learning model for relations between aspect word dependency types and aspect word relationships. To tackle the challenge of distinguishing sentiment across multiple aspects, the model employs a Simplified Type-Aware Graph Convolutional Network (ST-GCN) with multiple hierarchical layers to enhance ABSA. Specifically, the model integrates the contextual dependencies of words with their dependency types, enabling a comprehensive learning of dependency relationships. This allows the model to effectively differentiate between various aspect terms. An attention mechanism is applied to the dependency tree, where each edge is weighted according to its importance in the task, ensuring that significant dependencies receive more attention. Finally, context information learned from different GCN layers is combined using an attention layer set, ensuring that the sentiment is correctly associated with the relevant aspect term, thus avoiding confusion or noise in multi-aspect scenarios. To address

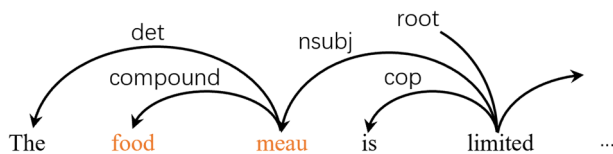


Fig. 1 An example sentence(including the aspect term “food menu”) with its dependency parsing

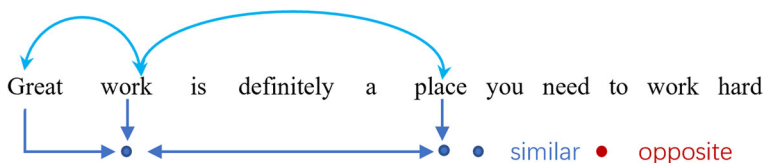


Fig. 2 Example of Inter-aspect relations

the challenge of handling long-distance dependencies, the model constructs a graph based on the dependency parsing results of the input text, leveraging GCN to capture long-range dependencies between words. The dependency tree is constructed using pre-existing parsing tools, and each edge in the graph is labeled according to the dependency type between the connected words. This graph structure allows the model to effectively capture long-distance word dependencies, mitigating the difficulties posed by long-range relationships in sentiment prediction. Additionally, the attention mechanism is applied to the graph, with all edges being weighted based on their contribution to the task. This ensures that important long-distance dependencies are properly emphasized, ultimately improving the model’s ability to handle complex dependency structures.

Our contributions can be summarized as:

- We propose a Simplified Type-Aware Graph Convolutional Network (T-GCN) for Multi-Aspect Sentiment Differentiation. The model enhances ABSA by integrating contextual dependencies and dependency types using a hierarchical ST-GCN.
- We introduce an attention mechanism that prioritizes key dependencies, ensuring accurate sentiment association with specific aspect terms, particularly in multi-aspect scenarios.
- We propose a convolutional network called Rely and Relation GCN (RR-GCN), which constructs a graph of the relationship between dependency types and aspect words.
- Comprehensive experiments on multiple challenging datasets demonstrate that the proposed method achieves state-of-the-art performance on the ABSA task.

2 Related Work

2.1 Aspect-Based Sentiment Analysis Systems

Traditional sentiment analysis approaches often rely on training classifiers using features like bag-of-words and sentiment lexicons [19]. However, both rule-based methods [20] and statistical approaches [21] depend heavily on handcrafted features, highlighting their limitations, especially when dealing with the complexity of customer review analysis. These methods struggle to effectively capture subtle, aspect-specific sentiments, underscoring the need for more sophisticated techniques.

Due to the limitations of traditional sentiment analysis, there has been a growing shift towards aspect-based sentiment analysis (ABSA). ABSA encompasses four primary tasks [21]: aspect term sentiment analysis (ATSA) [5], aspect category sentiment analysis (ACSA) [22], aspect term extraction (ATE) [23], and aspect category extraction (ACE) [24]. ATSA focuses on predicting the sentiment polarity of specified aspect terms, providing a more detailed understanding of sentiment expressions tied to specific elements in reviews [25]. ACSA, on the other hand, predicts sentiment polarity for predefined categories, offering insights into broader sentiment patterns related to particular aspects [26]. ATE and ACE, framed as entity extraction tasks, address the challenge of identifying and interpreting aspects within reviews. ATE targets the extraction of precise aspect terms, such as “dish,” aiming to pinpoint specific entities of interest [27]. ACE, meanwhile, identifies and categorizes broader aspects, like the “food” category, which includes terms such as “dish”.

The motivation behind these tasks is to attain a more nuanced and focused understanding of sentiment expressions across various contexts. By honing in on sentiments linked to specific aspects, aspect-based sentiment analysis enhances both the accuracy and relevance of sentiment analysis [28]. Nonetheless, challenges remain, particularly in developing robust methods to manage ambiguous language and contextual subtleties, highlighting the need for further research to improve the effectiveness of ABSA techniques.

2.2 RNN-Based Sentiment Analysis Systems

Recent advances in neural networks have gained significant attention for various applications [28–30]. The target-dependent sentiment classification model proposed in [31] leverages neural networks, specifically utilizing LSTM for aspect-based sentiment analysis. Building on LSTM, the target-dependent LSTM (TD-LSTM) incorporates target-specific information, effectively capturing the interaction between target words and their context to determine sentiment polarity. The model is trained end-to-end via backpropagation, using cross-entropy loss for supervised sentiment classification. However, while TD-LSTM improves sentiment classification by focusing on target-specific details, capturing subtle nuances and context-specific variations remains a challenge.

Similar to TD-LSTM, ATAE-LSTM [12] is designed to emphasize contextual relationships, employing both Bi-LSTM and attention-based LSTM. While these models have strengths, they struggle to capture long-range contextual information, particularly when key details are distant from the target. To address this, RAM [6] and TNet-LF [32] extend ATAE-LSTM [12] by incorporating multi-attention and Bi-attention mechanisms, respectively, improving the ability to handle long-range dependencies. However, TNet-LF still falls short in capturing complex sentence structures and grammar, limiting its ability to fully understand nuanced syntactic relationships. Moreover, its reliance on semantic aspects may lead to suboptimal performance when dealing with intricate syntactic constructs. Sentic LSTM [33] introduces a more advanced approach by integrating commonsense knowledge into an attentive LSTM network for targeted aspect-based sentiment analysis. Additionally, RACL [34], a relation-aware collaborative learning system, addresses key tasks like aspect term extraction, opinion term extraction, and aspect-level sentiment classification. However, the lack of comprehensive evaluations against other models limits a full understanding of RACL’s real-world performance.

The Transformer [35], a groundbreaking architecture, has proven highly effective in capturing complex long-range dependencies in textual data. Its ability to handle extensive

contextual relationships makes it a strong candidate for tasks that require detailed and robust analysis of such dependencies.

2.3 GCN-Based Sentiment Analysis Systems

Some early work in aspect-based sentiment analysis (ABSA) primarily used machine learning algorithms to capture sentiment polarity based on features extracted from content and syntactic structures [36, 37]. More recently, inspired by the notable performance of Graph Convolutional Networks (GCNs) across various tasks [38–40], some researchers have also applied them to sentiment analysis tasks. For instance, Zhang et al. [41] utilized GCNs to capture the dependency between syntactic information and aspect words by applying it to the sentence's dependency tree, yielding strong results. Sun et al. [42] proposed a GCN model that enhances aspect representation learned by Bi-LSTM through the sentence's dependency tree, while Tang et al. [43] introduced a dependency graph-enhanced bidirectional Transformer network (DGEDRT). Liang et al. [44] combined constituent syntactic trees and neural networks, using their hierarchical structure to organize links between different aspects for better sentiment polarity determination. Zhang et al. [45] applied a dynamic weighting mechanism to reweigh important words, improving semantic understanding. Further advancements in GCN-based ABSA include models like ASGCN [46], which captures relationships between aspects and words but relies heavily on accurate dependency tree parsing. AEGCN [47] integrates GCNs with multi-head attention, though its effectiveness depends on precise attention encoding. SK-GCN [48] combines syntactic dependency trees with GCNs but faces challenges in representing commonsense knowledge, while AGCN [49] aggregates node features, though its performance is limited by the completeness of the graph structure.

3 Methodology

As shown in Fig. 3, the architecture of the proposed RR-GCN model contains four main components. (1) word embeddings; (2) a dependency type graph convolutional network designed to capture dependencies using dependency types; (3) an aspect inter-word relationship extraction module designed to derive hidden vectors for different aspects; and (4) an attention score for fusing the two feature representations for the determination of aspect-specific sentiment features. The model will be described in the following.

3.1 Task Definition and Annotation

This chapter defines two tasks: type-aware utilization task and aspect word vector acquisition task.

3.1.1 Type-Aware Graph Convolutional Networks

Given a sentence $S = [w_1, w_2, \dots, w_n]$ with n words and m aspects $T = [A_1, A_2, \dots, A_m]$, where the i -th aspect $A_i = [w_j, w_{j+1} \dots w_k]$ is the subsequence of S . The task is to use a type-aware graph convolutional network (T-GCN) with multiple layers to enhance ABSA by incorporating both word relations and their dependency types to comprehensively learn from dependency parsing results.

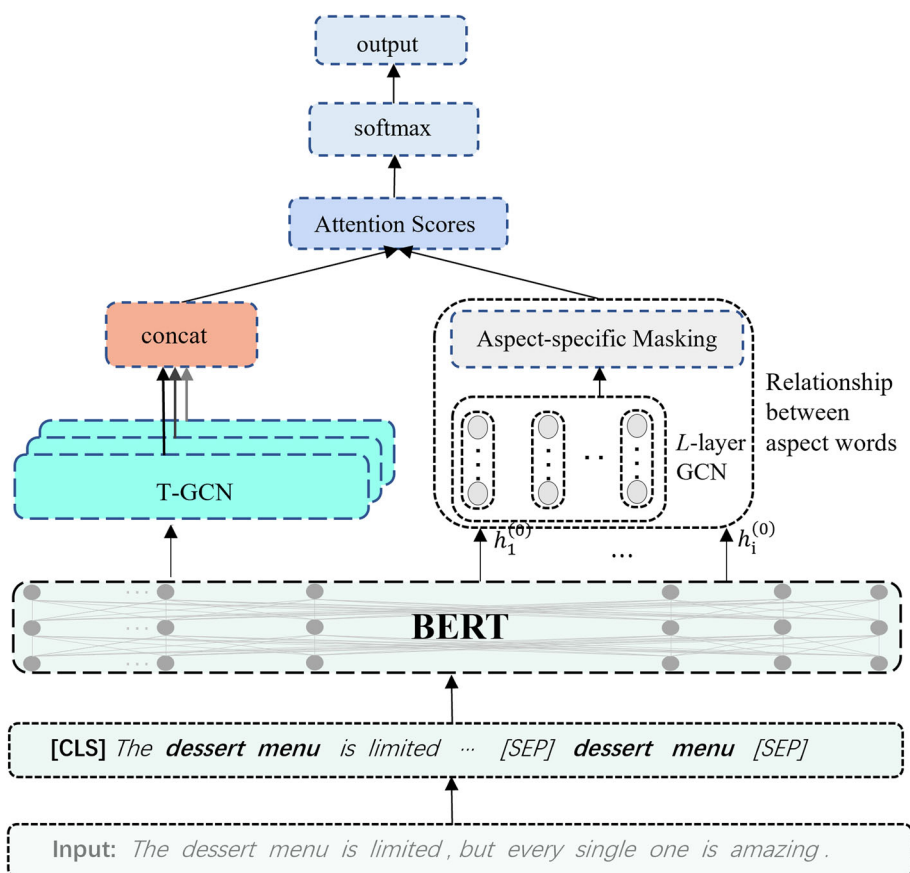


Fig. 3 An illustration of how we build the Jointly Learning Model

3.1.2 Aspect Vector Acquisition Task

For this task, there are usually three relations defined between aspect words, i.e., “similar”, “irrelevant” and “opposite”, as shown in Table 1. The relation labels can be automatically labeled according to the cosine similarity of the two aspects. Still, in order to make fuller use of the hidden vectors, this chapter outputs the aspect word vectors directly into the square attention for learning.

3.2 Embedding Layer

The embedding layer maps words into low-dimensional vectors, with $BERT_{base}$ used to obtain word embeddings for both aspect words and contexts. As shown in Fig. 3, the input format is $[[CLS], S, [SEP], a]$, where S represents the context and a is the aspect word. $[CLS]$ indicates a classification task, and $[SEP]$ separates sentences, corresponding to the last word’s vector, e.g., “dessert menu.” After vectorization, X is generated, and $BERT$ produces the word vector $H^x \in \mathbb{R}^{d_w}$, where d_w is the word vector dimension.

$$H^x = BERT(X) \quad (1)$$

3.3 Type-Aware Graph Construction

T-GCN is constructed on the basis of dependency types, so it is necessary to first focus on how to construct dependency types. Before constructing dependency types, syntactic dependencies need to be obtained, and dependency results need to be obtained using syntactic dependency analysis generics and Stanford converters. The results can be represented as a list of dependency tuples $(x_i, x_j, r_{i,j})$, with $r_{i,j}$ denoting the dependency type between x_i and x_j . Secondly, an adjacency matrix $\mathbf{A} = \{a_{i,j}\}_{n \times n}$ is used to represent edges with dependent types. Thus, $\mathbf{A}$ is a 0 and 1 matrix where $a_{i,j} = 1$ if there is an edge between x_i and x_j , and $a_{i,j} = 0$ otherwise. For $\mathbf{R}$, each of the elements $r_{i,j}$ (r_0 to r_5 in that order is represented by the type “none”, “self”, “det”, “compound”, “nsubj” and “cop”) with a token to indicate the type of dependency between x_i and x_j . Finally, to utilize the relationship types, a transition matrix is used to map all $r_{i,j}$ to their embeddings $e_{i,j}^r$. x_i and $e_{i,j}^r$ embeddings are connected by the following formula (where $\mathbf{h}_i^{(0)}$ is used to encode them in context).

$$\mathbf{s}_i^{(l)} = \mathbf{h}_i^{(l-1)} \oplus \mathbf{e}_{i,j}^r \quad (2)$$

$$\mathbf{s}_j^{(l)} = \mathbf{h}_j^{(l-1)} \oplus \mathbf{e}_{i,j}^r \quad (3)$$

The dependency type embedding schematic is shown in Fig. 4. Then, we compute the weight $p_{i,j}^{(l)}$ for this edge by

$$p_{i,j}^{(l)} = \frac{a_{i,j} \cdot \exp(\mathbf{s}_i^{(l)} \cdot \mathbf{s}_j^{(l)})}{\sum_{j=1}^n a_{i,j} \cdot \exp(\mathbf{s}_i^{(l)} \cdot \mathbf{s}_j^{(l)})} \quad (4)$$

The dimension of $e_{i,j}^r$ is aligned to the dimension of $\mathbf{h}_i^{(l-1)}$ through the trainable matrix $\mathbf{W}_R^{(l)}$ of the l -layer GCN by the following formula:

$$\mathbf{h}_j^{(l-1)'} = \mathbf{h}_j^{(l-1)} + \mathbf{W}_R^{(l)} \cdot \mathbf{e}_{i,j}^r \quad (5)$$

Finally, $p_{i,j}^{(l)}$ is applied to this edge according to the following formulae to compute the output of x_i in layer l following a similar process in traditional GCNs:

$$\mathbf{h}_i^{(l)} = \sigma \left(\sum_{j=1}^n p_{ij} \left(\mathbf{W}^{(l)} \cdot \mathbf{h}_j^{(l-1)'} + \mathbf{b}^{(l)} \right) \right) \quad (6)$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ denote trainable parameters in the l -th GCN layer and σ refer to the ReLU activation function.

For each word x_i , multiple T-GCN layers can learn indirect lexical relations at a distance since each T-GCN layer integrates information about the words directly connected to it. Therefore, this chapter assumes that different layers have the unique ability to encode contextual information. In order to exploit this ability, this chapter proposes the use of attentional layer integration to comprehensively learn the output of all T-GCN layers. To this end, the

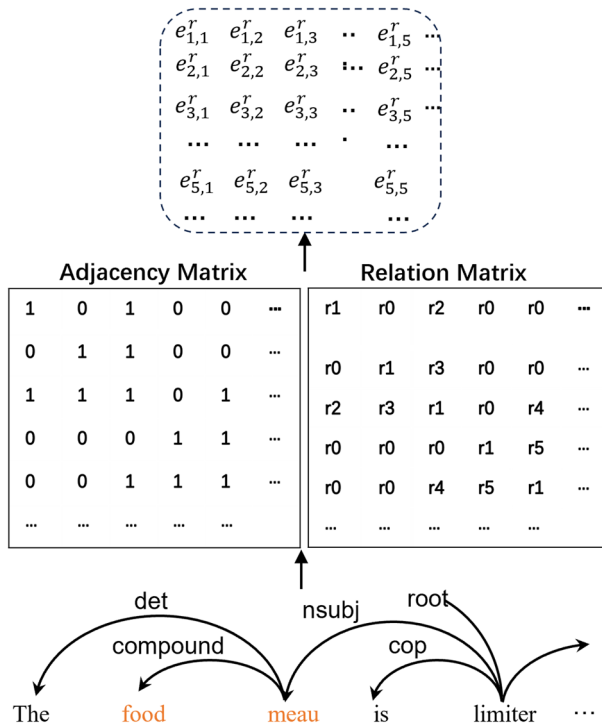


Fig. 4 The process of building the type-aware graph

output of each T-GCN layer $\mathbf{o}^{(l)}$ is first obtained by averaging the output hidden vectors of all aspect terms $x_k \in \mathcal{A}$:

$$\mathbf{o}^{(l)} = \frac{1}{|\mathcal{A}|} \cdot \sum_{x_k \in \mathcal{A}} h_k^{(l)} \quad (7)$$

where $|\mathcal{A}|$ denotes the number of aspect terms in the aspect terms $\mathcal{A}$. This chapter then pools the outputs of all T-GCN layers by weighted averaging:

$$\mathbf{o} = \sum_{l=1}^L \delta^{(l)} \cdot \mathbf{o}^{(l)} \quad (8)$$

3.4 Aspect Inter-Word Relationship Extraction Module

Due to its ability to process graph-structured data, GCN performs well in obtaining neighborhood information. As shown in Fig. 4, given a dependency graph on a sentence, the corresponding undirected neighbor matrix $A \in R^{n \times n}$ can be obtained. In a layer of GCN, the i th node h_i is updated in the following way:

$$\begin{aligned}
 h_i^{(1)} &= \text{ReLU} \left(\sum_{j=1}^n (A_{ij} + I_{ij}) W h_j + b \right) \\
 &= \text{ReLU} \left(\sum_{j=1}^n A_{ij} W h_j + W h_i + b \right)
 \end{aligned} \tag{9}$$

where $W \in \mathbb{R}^{2d' \times 2d'}$ is weight matrix, $b \in \mathbb{R}^{2d'}$ is a bias term, and $I \in \mathbb{R}^{n \times n}$ is the identity matrix considering self-connections. With L-layer GCNs, features from L hops away can be propagated to a target node. In this way, the syntactically relevant words can be transformed through the dependency graph-based GCNs. The representation of the i -th node in the l -th layer can be denoted as:

$$h_i^{(l)} = \text{ReLU} \left(\sum_{j=1}^n A_{ij} W^{l-1} h_j^{l-1} + W h_i^{l-1} + b^{l-1} \right) \tag{10}$$

based on h_i^l , the i -th aspect representation can be obtained through its original sequence index:

$$A_i = [h_j^l, h_{j+1}^l, \dots, h_k^l], \quad 0 \leq j \leq k \leq n \tag{11}$$

where j and k are the starting and ending indexes of the aspect.

3.5 Output Layer

For decoding, this chapter combines the dependency type relation $R_{i,j}$ and the relation H^l between aspect words using attention scores.

$$a = \text{softmax} \left(\frac{o \cdot (H^l)^T}{\sqrt{d_k}} \right) H^l \tag{12}$$

Then o is mapped to the labeling space via a fully connected layer, $u = W \bullet a + b$, where W and b are the trainable matrices and biases. y represents the polarity of the final predicted sentiment.

$$y = \text{softmax}(u) \tag{13}$$

4 Experiments

4.1 Datasets and Parameters

Our experiments are conducted on six public ABSA datasets: LAP14 and REST14 from Pontiki et al.(2014), REST15 from Pontiki et al.(2015), REST16 from Pontiki et al.(2016), and TWITTER from Dong et al. The detailed distributions of five datasets are presented in Table 1.

Table 1 Statistic of datasets

Dataset	Positive Train	Test	Neutral Train	Test	Negative Train	Test
LAPI4	994	341	464	169	870	128
REST14	2164	728	637	196	807	196
REST15	1178	439	50	35	382	328
REST16	1620	597	88	38	709	190
TWITTER	1561	173	3127	346	1560	173

The pre-trained *Bert_{base}* [50, 51] are used for word embedding initialization. The dimension of embedding and hidden states d_h is 768. Following standard methods, we tune our model using fivefold validation and grid-searching on the training set. The learning rate lr is selected from 0.1, 0.001, 0.0005, 0.0001, the batch size $b \in \{32, 64\}$, and the hidden size $d \in \{64, 128, 256, 768\}$. Adam is adopted to minimize the total loss L given in Eq. The regularization weight of parameters is 10^{-5} , the dropout rate is 0.3, the epoch number is 20, and the final parameters are presented in T The dropout rate is 0.1 to avoid overfitting. To evaluate the performance of ABSA models, Accuracy and Macro-Averaged F1 are adopted as the evaluation metrics. Our MWM-GCN is trained on RTX 3090 GPU, Intel i9-10900x CPU with PyTorch 1.9.0, and it costs 26h to train the model.

4.2 Concrete Implementation

Firstly, in order to build the graph of T-GCN, this chapter uses the best current component syntactic analyzer SPar3 to parse the input text into component trees, and then converts the trees into dependency trees via the Stanford Converter and finally builds the graph of dependencies and types from the trees. In this chapter, BERT is used as a context encoder because a high-quality text representation can improve the performance of NLP models. For this model BERT-base was used with three T-GCN layers and two ways of encoding the input were tried: encoding individual sentences, or encoding sentence-aspect pairs. For all models, the pre-training parameters in this chapter were initialized using BERT, and the other trainable parameters were initialized using Xavier. In addition, the model uses a cross-entropy loss function and the model is evaluated by accuracy and macro-averaged F1 scores across all sentiment polarities, which is based on previous research.

4.3 Models for Comparison

In order to evaluate the performance of the models in this chapter, representative models are selected as baseline models and compared in this subsection. There are two categories and the difference is whether they are based on GCN or not.

The GCN-based baseline model is as follows:

- *ASGCN* Zhang et al. [46] capture syntactic information and word dependencies via graph convolutional networks.
- *R-GAT* Wang et al. [52] construct a specific dependency tree rooted in aspectual terms by reshaping the dependency resolution tree encoded in a relational graph attention network.

- *DGEDT* Tang et al. [43] design a dual-transformer structure to achieve enhanced interaction between the planar representations learned from the transformer and the graph-based representations learned from the corresponding dependency tree.
- *DualGCN* Li et al. [53] use dual graph convolutional networks, i.e., SynGCN and SemGCN, considering both the complementary nature of syntactic structure and semantic association.
- *T-GCN* Tian et al. [54] use type-aware graph convolutional networks for dependency types and use attention mechanisms in T-GCN to distinguish different edges in the graph.
- *AGCN* [55] is an AGCN model that uses BERT to encode sentences.
- *Sentic GCN* [56] is a Sentic GCN model that uses BERT to encode sentences.

Other baseline models are shown below:

- *CDT* Sun et al. [42] combine dependency trees with neural networks to solve aspect-level sentiment analysis problems, BiLSTM is used to learn feature representations of sentences, and then GCN is used to augment the word embeddings learned from BiLSTM.
- *PBAN* Li et al. [57] design a neural network model based on the attention mechanism. The model combines polar and aspectual attention and is able to capture the polar and aspectual information of the text separately and combine the two for the judgment of emotional tendency.
- *RGAT* [58] proposes a relational graph attention network to model different types of dependencies and obtain syntax information. In addition, it adopts BERT to encode sentences to learn context information.
- *LTNMR* [59] is an LTNMR model that uses BERT to encode sentences.
- *RMN* [60] is an RMN model that uses BERT to encode sentences.
- *HGCN* Xu et al. [61] employ a range concept to learn structural ranges and introduce a hybrid convolutional network. This network fuses information from constituency and dependency trees to create knowledge-rich vector representations.
- *MT-BERT* Zhao et al. [62] introduce a multitask learning model that leverages aspect term extraction to enhance aspect polarity classification. A multi-head attention mechanism is utilized to capture the relevant dependency sequences.

Firstly, the performance of GCN-based models is generally 2% higher than other baseline models, which directly proves that GCN is excellent for feature extraction. Then each GCN-based baseline model is analyzed. ASGCN, although a hierarchical attention pooling strategy is used to enhance attention, the semantic information utilized is more homogeneous, so the effect is not satisfactory. Although R-GAT is also based on the attention mechanism, it introduces syntactic information, so the model effect has more than 3% improvement relative to ASGCN. DGEDT is again slightly better because it incorporates both transformer and features extracted from the dependency tree. DualGCN and T-GCN have the best results among several models. Compared to DualGCN, T-GCN has more influential dependency types, fuses the dependency information perfectly using dependency type graph convolution, and weights features at different levels using layer set attention, thus achieving better results in these baseline models.

CDT uses a combination of dependency trees and neural networks to achieve better results. PBAN, on the other hand, makes full use of the attention mechanism by combining polar attention and aspectual attention and fuses the two to achieve extreme results, almost to the effect of the GCN-based model.

As can be seen from Table 2, the RR-GCN model outperforms the GCN-based and attention-based models by roughly 1% performance improvement, indicating the limited

Table 2 Performance comparison on different methods on benchmark datasets

Dataset	LAP14		REST14		REST15		REST16		TWITTER	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
ASGCN	75.55	71.05	72.15	70.40	80.77	72.02	79.89	61.89	88.99	67.48
R-GAT	78.21	74.07	76.15	74.88	86.60	81.35	–	–	–	–
DGEDT	79.80	75.60	77.90	75.40	86.30	80.00	84.00	71	91.9	79
DualGCN	81.8	78.1	77.4	76.02	87.13	81.16	–	–	–	–
T-GCN	81.97	78.71	78.03	77.31	87.41	82.23	86	72.81	92.97	80.07
CDT	77.19	72.99	82.30	74.02	–	–	85.58	69.93	74.66	76.60
PBAN	79.80	75.60	86.30	80.00	84.00	71.00	91.90	79.00	77.90	75.40
RGAT	80.94	78.20	86.68	80.92	–	–	–	–	76.28	75.25
LTNMR	79.77	76.51	85.36	80.96	–	–	–	–	76.02	73.34
AGCN	79.94	76.52	82.77	73.29	82.84	65.08	88.80	67.65	75.43	74.11
RMN	77.95	70.83	84.56	79.05	82.94	66.95	89.38	71.88	–	–
Sentic GCN	82.12	79.05	86.92	81.03	85.32	71.28	91.97	79.56	–	–
HGCN	79.59	76.24	86.45	80.60	–	–	–	–	76.52	75.37
MT-BERT	80.56	77.00	86.88	81.16	–	–	–	–	76.59	74.47
Our model	81.63	78.39	87.21	82.1	85.8	72.71	92.8	80.01	77.93	77.19

effect of relationship assistance between different aspects of words. In contrast to models such as CDT and PBAN, RR-GCN uses syntactic information to introduce features that depend on type relationships, avoiding the noise that is introduced when using attention in some cases. Moreover, RR-GCN obtains better performance than GCN-based methods. On the one hand, the model itself partially uses GCN, and on the other hand, a clever combination of relationships between aspectual words aids in the judgment of sentiment polarity. Compared to the baseline GCN model, the RR-GCN model improves on both ACC and F1 by about another 1%. It is not difficult to see that the fusion of the use of dependency type relations and relations between aspect words increases the hidden representation of context, and more feature information, and improves the effect of aspect-level sentiment analysis judgement.

5 Ablation Studies

In order to explore the effectiveness of the different components of the model, i.e., the type-aware graph (TG), the attention score (AS), and the non-aspect word masking (AM), this subsection conducts an ablation study based on the model. Table 3 reports the experimental results using different combinations of these components on all datasets:

First, the first (ID: 1) and last (ID: 8) rows illustrate the results of the baseline for the full model and the normal GCN, respectively. Here, the models without AM (ID: 4-6) use the output of the last T-GCN layer (i.e., the third layer) to predict the polarity of emotions. Firstly, the results clearly show that on all datasets, the performance of the model degrades if any component is excluded from the full model. This observation suggests that all three components play an important role in the approach of this chapter to enhance ABSA; each component has its own unique contribution to the overall model.

Table 3 Experimental results of ablation study on the five datasets, with different configurations applied to model for this chapter

Setting ID	LAP14			F1		REST14		REST15		REST16		TWITTER	
	TG	AS	AM	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
1	✓	✓	✓	81.22	77.97	86.43	80.23	85.93	72.71	92.57	77.41	77.35	75.30
2	✗	✓	✓	81.03	77.79	86.20	80.15	85.31	72.06	92.04	77.10	77.11	75.02
3	✓	✗	✓	80.58	77.56	86.24	80.19	85.64	71.03	91.19	76.70	76.63	75.10
4	✓	✓	✗	80.25	76.92	85.54	78.86	85.07	72.50	91.83	76.86	76.16	74.44
5	✓	✗	✗	80.19	76.32	85.48	78.32	84.73	70.50	90.65	75.54	75.85	74.68
6	✗	✓	✗	80.11	76.82	85.42	78.02	84.87	70.03	90.79	74.70	75.61	74.38
7	✗	✗	✓	79.65	76.86	85.31	78.11	84.78	70.21	90.46	74.62	75.63	74.57
8	✗	✗	✗	78.01	76.45	84.18	78.13	82.42	69.32	87.39	71.40	71.61	72.12

In the table, “ID” stands for the number of rows, “TG” refers to the type-perception graph; “AS” stands for attention score; “AM” represents non-aspect word masking. The “✓” and “✗” indicate whether the corresponding components are used or not

Second, for each single component, the results for models with specific modules (ID: 5–7) compared to the results for the GCN baseline (ID: 8) show that type-dependent graph convolution is the most important mechanism for improving model performance, with the model with type-dependent graph convolution (ID: 5) outperforming the others in the three cases where a single module is used. This observation is in line with our intuition since type-dependent graph convolution is the process of integrating the dependency type information well into the graph convolution network and weighting the different contributions of features at different levels to maximise the contribution of the type-dependent information features. This also shows that being able to extract higher-level features is the most redundant factor. Next is the attention mechanism, as attention directly guides the model to differentiate between contextual information and aspectual words, reducing the noise introduced by the inherent nature of the dependency tree structure, thus enabling informative words to be highlighted, and thus improving the ABSA accordingly. whereas the relationship of aspectual words aids the other important features to function only when they are used for polarity determination and is ineffective when acting alone.

6 Effect of Different T-GCN Layers

In addition to these components, this chapter also investigates the effect of each layer when the model is trained on different datasets. In doing so, this subsection experiments with the best-performing models on all datasets and uses weights (for each T-GCN layer) to identify their contributions. The results are shown in Fig. 5, where the weights for T-GCN layers 1, 2, and 3 are blue, orange, and grey, respectively. There are the following final observations.

Firstly, all layers contribute to the final prediction of ABSA, which is in line with the expectations of this subsection and confirms the validity of utilizing information from all layers of the GCN. Thus, the model is able to provide comprehensive contextual information compared to using only the output of the last layer.

Second, as shown in the histograms, the second layer of the T-GCN contributes the most among all three layers on the five datasets. One possible reason for this is that (1) the second layer is able to encode a larger range of contextual information (since the edges of the first

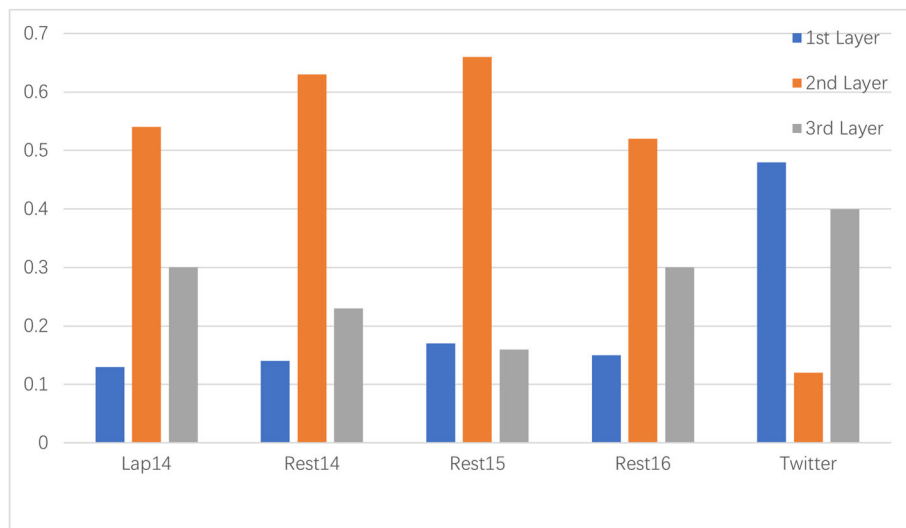


Fig. 5 The histograms of weights assigned to different T-GCN layers (blue, orange and gray bars represent the weights of the first, second, and third layers respectively.) in ALE with respect to each dataset

layer only cover words with direct relationships, whereas the second and third layers provide indirect relationships, i.e., second- and third-order dependencies in practice); and (2) the second layer is likely to introduce less irrelevant information from multi-word relationships compared to the third layer.

Third, this chapter also notes that for TWITTER, the distribution of weights among the three layers is rather different from the rest of the dataset, with the first and last layers contributing more to ABSA. This observation can be explained by the fact that TWITTER is social media data, and in general, sentences in this type of data are short and have little organization, as the models in this chapter may require information from local contexts or whole sentences for ABSA.

7 Conclusions

Based on the two features of dependency types and relations between aspect words, this paper constructs a new ABSA model and discusses its design ideas, construction, and details of comparison experiments. The model uses the T-GCN technique to extract the feature information of dependency types and designs a module for extracting the relationship between aspect words to obtain the hidden vectors of different aspect words. Finally, weight distribution self-learning is performed by attention scores. In the experiments, five public datasets are used to compare the model of this study with the classical ABSA model to ensure its effectiveness.

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